

Comparing Nested Models

STAT 223 – Applied Analytics

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Comparing Nested Models

When we discussed testing for goodness of fit and the significance of predictors, we discussed two scenarios: testing each predictor individually with a t -test and testing the full model all together using the F -test.

But there are times we want to do something in between.

Comparing Nested Models

Suppose we have a model with k predictors, and we have another set of $p - k$ predictors that we are considering adding to the model.

This creates a “nested” model:

$$\text{Reduced Model: } \hat{Y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon$$

$$\text{Full Model: } \hat{Y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \beta_{k+1} x_{k+1} + \cdots + \beta_p x_p + \varepsilon$$

We can test if the “full” model is a significant improvement over the “reduced” model with an F -test.

F Distribution

- Let $Y_i \stackrel{ind}{\sim} \text{NORMAL}(\mu_i, \sigma_i^2)$.
- Let $Z_i = \frac{Y_i - \mu_i}{\sigma_i}$; $Z_i \stackrel{iid}{\sim} \text{NORMAL}(0, 1)$.
- Let $Q_1 = Z_1^2 + \cdots + Z_{n_1}^2$ and $Q_2 = Z_{n_1+1}^2 + \cdots + Z_{n_1+n_2}^2$.
- Then $Q_1 \sim \chi_{n_1}^2$, with mean n_1 and variance $2n_1$ and $Q_2 \sim \chi_{n_2}^2$, with mean n_2 and variance $2n_2$.
- Q_1 is independent of Q_2 .
- Then:

$$F = \frac{Q_1/n_1}{Q_2/n_2} \sim F_{n_1, n_2}$$

- This is an F distribution with $df_1 = n_1$ numerator degrees of freedom and $df_2 = n_2$ denominator degrees of freedom.

Using the F distribution, we can develop a test to compare nested models:

$$H_0 : \beta_{k+1} = \beta_{k+2} = \cdots = \beta_p = 0$$

$$H_A : \text{At least one of the } \beta_i \text{'s are non-zero for } i \in \{k+1, k+2, \dots, p\}$$

Then the test statistic is:

$$F = \frac{(SSR - SSE)/p-k}{SSE/n-p-1} \sim F_{p-k, n-p-1}$$

Likelihood Ratio Test

The F test is a special case of a more general procedure for comparing nested model: the likelihood ratio test:

$$\chi^2 = -2 \left(\ln(L_{reduced}) - \ln(L_{Full}) \right) \sim \chi^2_{p-k},$$

where L is the likelihood function, which measures how probable it is that the observed data was generated by that specific model.

The F test and the likelihood ratio test can only be applied to nested models.

For non-nested models, information criteria can be used.

Akaike's Information Criterion (AIC)

Akaike's Information Criterion (AIC), where K is the number of parameters of model M :

$$AIC = 2 \ln(L_M) + 2K$$

The AIC measures the distance between the probability distribution specified by model M and the (hypothetical) true distribution.

Bayesian Information Criterion (BIC)

An alternative to the AIC is the **Bayesian Information Criterion (BIC)**, with N the number of data points used to calculate the likelihood:

$$BIC = -2 \ln(L_M) + K \ln N.$$

Based on the BIC, we can compare two models M_1 and M_2 using the *Bayes factor*:

$$B = \exp \left(-\frac{1}{2} \Delta BIC \right),$$

where $\Delta BIC = BIC_{M_1} - BIC_{M_2}$.

Formally, the Bayes factor measures the ratio of the posterior probabilities of the two models, $P(M_1|y)/P(M_2|y)$.